

Supplementary Material for “Structural Labels for Stratified Audit Budget Allocation in Knowledge-Graph Dependency Flows”

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1 Appendix A. Synthetic scalability and threshold sensitivity

Appendix A supports scalability beyond the Countries-KG semantic fixture without claiming robustness to arbitrary noisy open-domain KGs. The left anchor is the Countries-KG 30-node semantic fixture. Synthetic directed dependency graphs use seed 20260711 and sizes 100, 200, 500, 1000, 5000.

Table 1 Synthetic scalability curve with Countries-KG as the 30-node left anchor.

Nodes	Source	Macro F1
30	Countries-KG	1.000
100	synthetic_dag	1.000
200	synthetic_dag	1.000
500	synthetic_dag	1.000
1000	synthetic_dag	1.000
5000	synthetic_dag	1.000

The redundancy threshold sensitivity check compares $\theta = 0.85$ and $\theta = 0.90$. Countries-KG gives 124 redundancy positives at $\theta = 0.85$ and 124 positives at $\theta = 0.90$. The synthetic planted-flow runs remain finite through 5k nodes under both thresholds.

Table 2 Synthetic graph diagnostics, including Average Path Length to Covered Descendants.

Nodes	Macro F1	Average path length
100	1.000	2.587
200	1.000	2.819
500	1.000	3.115
1000	1.000	3.349
5000	1.000	3.883

2 Appendix B. Necessary Condition Diagnosis

This appendix uses `scripts/run_prm800k_strong_baselines.py` only as boundary evidence. It is titled Necessary Condition Diagnosis because the PRM800K route lacks the directed KG dependency-flow condition used in the main experiment.

Table 3 PRM800K necessary-condition context. The graph-only row is boundary evidence, not a main result.

Method	Spearman ρ	Mass@25%
w_struct	0.611	0.381
Structure graph + position	0.603	0.380
Structure graph only	0.043	0.311
Raw local utility	-0.077	0.282

The requested condition, Trace Length > 20 and graph density < 0.05 , is represented by an archived high-trace-length sparse-TF-IDF proxy because the locked report does not store an exact joint density stratum. Under that proxy, graph-only $\rho = 0.000$. This negative result defines the framework boundary: structural labels need directed dependency edges with at least moderate logical density.

3 Appendix C. Cached-label audit reproducibility

The audit case loads `outputs\countries_kg_label_validation\countries_kg_labels_cached.json`. The report records `recomputed_label_count=0`, seed 20260711, 600 simulated audit repetitions, and budget fraction 0.25. The cache includes node identifiers, bottleneck flags, redundancy flags, redundancy group identifiers, downstream impact counts, thresholds, random seed, KG metadata hash, and synthetic DAG configuration.

References